

RADHIKA DOTIHAL

San Jose, CA | (628) 363-7138 | radhikardotihal@gmail.com | github.com/raddot | linkedin.com/in/radhika-dotihal | raddot.github.io
Authorized to work in the U.S. (H-4 EAD)

SUMMARY

Backend software engineer with 5+ years building low-latency, high-throughput systems for production fintech environments. Redesigned a core C++ report engine to cut processing time 83% for 5,000+ clients. Extends backend expertise into AI-powered systems, including a Kubernetes-deployed RAG backend with vector search. Strong foundation in distributed systems, REST APIs, and cloud infrastructure (AWS, GCP).

EXPERIENCE

Java Instructor, TechElevate

Present

- Taught 30 students Java fundamentals (OOP, data structures) through bi-weekly sessions; created 10+ hands-on coding exercises

Software Engineer — SS&C Technologies, Inc. (Advent)

3 years 1

month

- Redesigned the core C++ report engine with multi-threaded, batched I/O, cutting report generation from 18 minutes to under 3 minutes (83% reduction) and enabling same-day delivery for 5,000+ production clients.
- Collaborated in a cross-functional team of 3 to automate DEV/PROD environment validation, eliminating a 3-hour manual setup across 20–25 monthly environment builds and saving 60–70 engineer-hours per month.
- Built a Python diagnostic tool to detect configuration divergence across environments, cutting debugging time 75% (2+ hours to under 40 minutes) and eliminating a recurring class of production incidents.
- Refactored legacy C++ report engine and introduced automated testing with JUnit, improving system stability for 5,000+ clients; collaborating cross-functionally across full SDLC/Agile cycles.

Front Desk Student Assistant — SUNY Binghamton — Office of International Student & Scholar Services

11 months

- Supported international student services office with administrative coordination and event logistics.
- Built an ARIMA demand-forecasting model for e-commerce sales data and evaluated accuracy using RMSE.
- Analyzed large datasets using K-Means clustering and MapReduce, visualizing results via pie and bar charts.

Volunteer Backend Engineer, Global Health Impact Project

4 months

- Built the backend (Python, Flask, SQLite, REST APIs) for a pharma-impact forecasting platform, still in production.

Software Engineer — Larsen & Toubro Infotech, India

2 years 3

months

- Orchestrated ETL migration across UAT/DEV/QA environments using Informatica PowerCenter, reducing release cycle defects ~80% across 80–110 deployments per quarter.
- Migrated an OBIEE BI repository serving ~300 users across Oracle BI Admin, Enterprise Manager, and Data Integrator with uninterrupted analysis access.
- Designed backend services in MongoDB with RESTful API routes; applied Template Method, Decorator, Observer, and Factory design patterns across feature work.
- Built a full-stack seat-booking application (Java, MySQL, JavaScript, PHP) serving 30–40 users, reducing support tickets 20% through improved input validation and error handling.
- Implemented a Test-Driven Development workflow with Mocha/Chai for an eateries-lookup backend service.

Software Engineer Intern — Greenvity Communications, India

6 months

- Built smart-home device control features (power, brightness, scheduling, zones) using AWS Lambda; trained Alexa intents/utterances in JSON and maintained the backing XML database.

PROJECTS

KubeRAG | Python, PostgreSQL, FastAPI, Redis, Docker, Kubernetes, Prometheus, Grafana [Project Link](#)

- Developed full-stack web application development with FastAPI enabling PDF upload and context-aware document Q&A, load tested with 200+ documents.
- Implemented async embedding generation and Redis caching for vector search to reduce query latency.

- Deployed microservices on Kubernetes (Docker Desktop) with Deployments, Services, ConfigMaps; configured liveness/readiness probes, scaling 3 -> 20 pods.
- Demonstrated strong problem-solving and attention to detail by debugging YAML indentation.
- Achieved 92% retrieval accuracy using cosine similarity, optimized chunking strategy which reduced LLM hallucinations by 60%.
- Instrumented RAG API with Prometheus metrics - latency, request count, error rate, ingest throughput exposing /metrics; configured Prometheus scraping every 5s, analyzed API performance metrics using Excel and built Grafana dashboards.

SKILLS

Languages: C++, Java, Python, JavaScript, Go, SQL, Bash, PHP

Backend & APIs: REST API, Express.js, Flask, FastAPI, Spring, JUnit

Databases: MySQL, MongoDB, SQLite, PostgreSQL, Redis

Cloud & DevOps: AWS (Lambda, DynamoDB), GCP, Docker, Kubernetes (K8s), Jenkins, CI/CD

AI & LLM Systems: RAG pipelines, vector search, embeddings, LangChain, AI-assisted development (Claude Code, Cursor)

Core Concepts: System Design, Data Structures & Algorithms, OOP, Design Patterns, Distributed Systems

Observability & Messaging: Terraform, Prometheus, Grafana, Apache Kafka

EDUCATION

Master of Science, Computer Science — State University of New York at Binghamton

Bachelor of Engineering, Information Technology — University of Pune, India

PUBLICATIONS & CERTIFICATIONS

- "Smart Homes Using Alexa and Power Line Communication in IoT" — presented and published at ICCCNT, Springer (Apr 2018).
- AI Agents, Chatbot Development, LLM Fundamentals & AI Ethics certification from SimpliLearn (Apr 2026).
- Self-paced coursework: Terraform, Grafana, Apache Kafka 101, Apache Kafka Connect 101 (May 2026).